Patient demographics tables for Balder ML modeling

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Overview

This document presents tables on patient demographics used for ML modeling. It primarily reports on GENIE data.

Loading data

Aggregation of data by cancer type

Data split by OncoTree codes

```
knitr::kable(outCntTbl, )
```

OT code	cancer type	cancer type detailed	number of patients	actionable events
LUAD	Lung Cancer	Lung Adenocarcinoma	18775	8019 (42.7%)
IDC	Breast Cancer	Breast Invasive Ductal Carcinoma	9282	3822 (41.2%)
BRCA	Breast Cancer	Invasive Breast Carcinoma	3141	1412 (45%)
GIST	Gastrointestinal Stromal Tumor	Gastrointestinal Stromal Tumor	1553	1212 (78%)
MEL	Melanoma	Melanoma	3671	1159 (31.6%)
COAD	Colorectal Cancer	Colon Adenocarcinoma	9647	1010 (10.5%)
THPA	Thyroid Cancer	Papillary Thyroid Cancer	1295	876 (67.6%)
ILC	Breast Cancer	Breast Invasive Lobular Carcinoma	1456	841 (57.8%)
PRAD	Prostate Cancer	Prostate Adenocarcinoma	5661	741 (13.1%)
AML	Acute myeloid leukemia	Acute myeloid leukemia	2167	701 (32.3%)
SKCM	Melanoma	Cutaneous Melanoma	1940	656 (33.8%)
BLCA	Bladder Cancer	Bladder Urothelial Carcinoma	3609	647 (17.9%)
NSCLC	Non-Small Cell Lung Cancer	Non-Small Cell Lung Cancer	1780	512 (28.8%)
ODG	Glioma	Oligodendroglioma	495	467 (94.3%)
HGSOC	Ovarian Cancer	High-Grade Serous Ovarian Cancer	3726	444 (11.9%)
ASTR	Glioma	Astrocytoma	577	368 (63.8%)
MDLC	Breast Cancer	Breast Mixed Ductal and Lobular Carcinoma	602	306 (50.8%)

Aggregation by sex and age

Data split by sex codes

```
knitr::kable(outSexTbl, )
```

cancer type detailed	male	female	uknown sex
Lung Adenocarcinoma	7331 (39%)	11369 (60.6%)	75 (0.4%)
Breast Invasive Ductal Carcinoma	78 (0.8%)	9148 (98.6%)	56 (0.6%)
Invasive Breast Carcinoma	30 (1%)	3100 (98.7%)	11 (0.4%)
Gastrointestinal Stromal Tumor	861 (55.4%)	686 (44.2%)	6 (0.4%)
Melanoma	2182 (59.4%)	1482 (40.4%)	7 (0.2%)
Colon Adenocarcinoma	5007 (51.9%)	4601 (47.7%)	39 (0.4%)
Papillary Thyroid Cancer	554 (42.8%)	740 (57.1%)	1 (0.1%)
Breast Invasive Lobular Carcinoma	0 (0%)	1445 (99.2%)	11 (0.8%)
Prostate Adenocarcinoma	5620 (99.3%)	4 (0.1%)	37 (0.7%)
Acute myeloid leukemia	1257 (58%)	902 (41.6%)	8 (0.4%)
Cutaneous Melanoma	1244 (64.1%)	692 (35.7%)	4 (0.2%)
Bladder Urothelial Carcinoma	2751 (76.2%)	848 (23.5%)	10 (0.3%)
Non-Small Cell Lung Cancer	845 (47.5%)	927 (52.1%)	8 (0.4%)
Oligodendroglioma	262 (52.9%)	233 (47.1%)	0 (0%)
High-Grade Serous Ovarian Cancer	0 (0%)	3715 (99.7%)	11 (0.3%)
Astrocytoma	345 (59.8%)	231 (40%)	1 (0.2%)
Breast Mixed Ductal and Lobular Carcinoma	4 (0.7%)	597 (99.2%)	1 (0.2%)

Aggregation by Ancestry

```
raceByEthn <- data.frame(table(perPatientLevel1$RACE,perPatientLevel1$ETHNICITY))
```

Most common mutated genes for each cancer type

top five actionable genes for each cancer type

```
knitr::kable(outGeneTbl, )
```

cancer type detailed	gene	count and perc of actionable events
Lung Adenocarcinoma	EGFR	4257 (53.1%)
Lung Adenocarcinoma	KRAS	2544 (31.7%)
Lung Adenocarcinoma	MET	466 (5.8%)
Lung Adenocarcinoma	ERBB2	436 (5.4%)
Lung Adenocarcinoma	BRAF	316 (3.9%)
Breast Invasive Ductal Carcinoma	PIK3CA	2773 (72.6%)
Breast Invasive Ductal Carcinoma	PTEN	440 (11.5%)
Breast Invasive Ductal Carcinoma	AKT1	351 (9.2%)
Breast Invasive Ductal Carcinoma	ESR1	255 (6.7%)
Breast Invasive Ductal Carcinoma	BRAF	3 (0.1%)
Invasive Breast Carcinoma	PIK3CA	869 (61.5%)
Invasive Breast Carcinoma	ESR1	241 (17.1%)
Invasive Breast Carcinoma	PTEN	164 (11.6%)
Invasive Breast Carcinoma	AKT1	135 (9.6%)
Invasive Breast Carcinoma	BRAF	3 (0.2%)
Gastrointestinal Stromal Tumor	KIT	1172 (96.7%)
Gastrointestinal Stromal Tumor	PDGFRA	34 (2.8%)
Gastrointestinal Stromal Tumor	BRAF	6 (0.5%)
Melanoma	BRAF	1159 (100%)
Colon Adenocarcinoma	BRAF	1010 (100%)
Papillary Thyroid Cancer	BRAF	876 (100%)
Breast Invasive Lobular Carcinoma	PIK3CA	619 (73.6%)
Breast Invasive Lobular Carcinoma	PTEN	85 (10.1%)
Breast Invasive Lobular Carcinoma	AKT1	68 (8.1%)
Breast Invasive Lobular Carcinoma	ESR1	67 (8%)
Breast Invasive Lobular Carcinoma	BRAF	2 (0.2%)
Prostate Adenocarcinoma	CDK12	231 (31.2%)
Prostate Adenocarcinoma	BRCA2	190 (25.6%)
Prostate Adenocarcinoma	ATM	167 (22.5%)
Prostate Adenocarcinoma	ATR	28 (3.8%)
Prostate Adenocarcinoma	CHEK2	20 (2.7%)
Acute myeloid leukemia	FLT3	273 (38.9%)
Acute myeloid leukemia	IDH2	269 (38.4%)
Acute myeloid leukemia	IDH1	159 (22.7%)
Cutaneous Melanoma	BRAF	656 (100%)
Bladder Urothelial Carcinoma	FGFR3	641 (99.1%)
Bladder Urothelial Carcinoma	BRAF	6 (0.9%)
Non-Small Cell Lung Cancer	KRAS	248 (48.4%)
Non-Small Cell Lung Cancer	EGFR	189 (36.9%)
Non-Small Cell Lung Cancer	MET	38 (7.4%)
Non-Small Cell Lung Cancer	ERBB2	22 (4.3%)
Non-Small Cell Lung Cancer	BRAF	15 (2.9%)
Oligodendroglioma	IDH1	415 (88.9%)
Oligodendroglioma	IDH2	52 (11.1%)
High-Grade Serous Ovarian Cancer	BRCA1	290 (65.3%)
High-Grade Serous Ovarian Cancer	BRCA2	152 (34.2%)
High-Grade Serous Ovarian Cancer	BRAF	2 (0.5%)
Astrocytoma	IDH1	327 (88.9%)
Astrocytoma	BRAF	30 (8.2%)
Astrocytoma	IDH2	11 (3%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	253 (82.7%)
Breast Mixed Ductal and Lobular Carcinoma	AKT1	23 (7.5%)
Breast Mixed Ductal and Lobular Carcinoma	PTEN	19 (6.2%)
Breast Mixed Ductal and Lobular Carcinoma	ESR1	10 (3.3%)
Breast Mixed Ductal and Lobular Carcinoma	BRAF	1 (0.3%)

Most coding mutations for each cancer type

Ten most common mutations for each cancer type

```
knitr::kable(outMutTbl, )
```

cancer type detailed	gene	protein mutation	count and perc of actionable events
Lung Adenocarcinoma	KRAS	p.G12C	2539 (31.7%)
Lung Adenocarcinoma	EGFR	p.L858R	1641 (20.5%)
Lung Adenocarcinoma	EGFR	p.E746_A750del	1329 (16.6%)
Lung Adenocarcinoma	BRAF	p.V600E	316 (3.9%)
Lung Adenocarcinoma	ERBB2	p.Y772_A775dup	238 (3%)
Lung Adenocarcinoma	MET	p.X1010_splice	190 (2.4%)
Lung Adenocarcinoma	EGFR	p.G719A	172 (2.1%)
Lung Adenocarcinoma	EGFR	p.L747_P753delinsS	150 (1.9%)
Lung Adenocarcinoma	EGFR	p.L861Q	148 (1.8%)
Lung Adenocarcinoma	EGFR	p.L747_T751del	108 (1.3%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.H1047R	1205 (31.5%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.E545K	617 (16.1%)
Breast Invasive Ductal Carcinoma	AKT1	p.E17K	351 (9.2%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.E542K	341 (8.9%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.N345K	185 (4.8%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.H1047L	162 (4.2%)
Breast Invasive Ductal Carcinoma	ESR1	p.D538G	101 (2.6%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.C420R	71 (1.9%)
Breast Invasive Ductal Carcinoma	ESR1	p.Y537S	69 (1.8%)
Breast Invasive Ductal Carcinoma	PIK3CA	p.G1049R	29 (0.8%)
Invasive Breast Carcinoma	PIK3CA	p.H1047R	346 (24.5%)
Invasive Breast Carcinoma	PIK3CA	p.E545K	203 (14.4%)
Invasive Breast Carcinoma	AKT1	p.E17K	135 (9.6%)
Invasive Breast Carcinoma	PIK3CA	p.E542K	135 (9.6%)
Invasive Breast Carcinoma	ESR1	p.D538G	104 (7.4%)
Invasive Breast Carcinoma	PIK3CA	p.N345K	58 (4.1%)
Invasive Breast Carcinoma	ESR1	p.Y537S	57 (4%)
Invasive Breast Carcinoma	PIK3CA	p.H1047L	45 (3.2%)
Invasive Breast Carcinoma	PIK3CA	p.C420R	27 (1.9%)
Invasive Breast Carcinoma	ESR1	p.Y537N	21 (1.5%)
Gastrointestinal Stromal Tumor	KIT	p.A502_Y503dup	124 (10.2%)
Gastrointestinal Stromal Tumor	KIT	p.W557_K558del	111 (9.2%)
Gastrointestinal Stromal Tumor	KIT	p.V559D	81 (6.7%)
Gastrointestinal Stromal Tumor	KIT	p.W557R	44 (3.6%)
Gastrointestinal Stromal Tumor	KIT	p.L576P	39 (3.2%)
Gastrointestinal Stromal Tumor	KIT	p.V654A	36 (3%)
Gastrointestinal Stromal Tumor	KIT	p.V560D	32 (2.6%)
Gastrointestinal Stromal Tumor	KIT	p.X550_splice	30 (2.5%)
Gastrointestinal Stromal Tumor	KIT	p.V559G	27 (2.2%)
Gastrointestinal Stromal Tumor	KIT	p.K642E	23 (1.9%)
Melanoma	BRAF	p.V600E	990 (85.4%)
Melanoma	BRAF	p.V600K	146 (12.6%)
Melanoma	BRAF	p.V600R	18 (1.6%)
Melanoma	BRAF	p.V600D	3 (0.3%)
Melanoma	BRAF	p.V600_K601delinsEN	1 (0.1%)
Melanoma	BRAF	p.V600_S602delinsRKT	1 (0.1%)
Colon Adenocarcinoma	BRAF	p.V600E	1009 (99.9%)
Colon Adenocarcinoma	BRAF	p.T599_V600delinsRE	1 (0.1%)
Papillary Thyroid Cancer	BRAF	p.V600E	876 (100%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.H1047R	234 (27.8%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.E545K	142 (16.9%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.E542K	98 (11.7%)
Breast Invasive Lobular Carcinoma	AKT1	p.E17K	68 (8.1%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.N345K	50 (5.9%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.H1047L	39 (4.6%)
Breast Invasive Lobular Carcinoma	ESR1	p.D538G	25 (3%)
Breast Invasive Lobular Carcinoma	ESR1	p.Y537S	18 (2.1%)
Breast Invasive Lobular Carcinoma	PIK3CA	p.C420R	18 (2.1%)
Breast Invasive Lobular Carcinoma	ESR1	p.E380Q	11 (1.3%)
Prostate Adenocarcinoma	BRCA2	p.W1692Mfs*3	8 (1.1%)
Prostate Adenocarcinoma	ATR	p.X1913_splice	6 (0.8%)
Prostate Adenocarcinoma	CDK12	p.Y319*	6 (0.8%)
Prostate Adenocarcinoma	ATM	p.S2812Vfs*3	4 (0.5%)
Prostate Adenocarcinoma	ATM	p.X2929_splice	4 (0.5%)
Prostate Adenocarcinoma	BRCA2	p.N1626Sfs*12	4 (0.5%)
Prostate Adenocarcinoma	CDK12	p.H224Pfs*6	4 (0.5%)
Prostate Adenocarcinoma	CDK12	p.K861Sfs*7	4 (0.5%)
Prostate Adenocarcinoma	CDK12	p.L123Tfs*4	4 (0.5%)
Prostate Adenocarcinoma	ATM	p.I323V	3 (0.4%)
Acute myeloid leukemia	IDH2	p.R140Q	176 (25.1%)
Acute myeloid leukemia	IDH1	p.R132C	93 (13.3%)
Acute myeloid leukemia	IDH2	p.R172K	74 (10.6%)
Acute myeloid leukemia	IDH1	p.R132H	43 (6.1%)
Acute myeloid leukemia	FLT3	p.D835Y	34 (4.9%)
Acute myeloid leukemia	FLT3	p.D593_F594insNS	20 (2.9%)

cancer type detailed	gene	protein mutation	count and perc of actionable events
Acute myeloid leukemia	FLT3	p.D835H	17 (2.4%)
Acute myeloid leukemia	IDH2	p.R140W	15 (2.1%)
Acute myeloid leukemia	IDH1	p.R132G	11 (1.6%)
Acute myeloid leukemia	FLT3	p.Y597_E598insDYVDFREY	10 (1.4%)
Cutaneous Melanoma	BRAF	p.V600E	550 (83.8%)
Cutaneous Melanoma	BRAF	p.V600K	82 (12.5%)
Cutaneous Melanoma	BRAF	p.V600R	16 (2.4%)
Cutaneous Melanoma	BRAF	p.V600D	4 (0.6%)
Cutaneous Melanoma	BRAF	p.V600G	3 (0.5%)
Cutaneous Melanoma	BRAF	p.V600_K601delinsEI	1 (0.2%)
Bladder Urothelial Carcinoma	FGFR3	p.S249C	425 (65.7%)
Bladder Urothelial Carcinoma	FGFR3	p.Y373C	113 (17.5%)
Bladder Urothelial Carcinoma	FGFR3	p.R248C	83 (12.8%)
Bladder Urothelial Carcinoma	FGFR3	p.G370C	19 (2.9%)
Bladder Urothelial Carcinoma	BRAF	p.V600E	6 (0.9%)
Bladder Urothelial Carcinoma	FGFR3	p.S371_Y373delinsGVC	1 (0.2%)
Non-Small Cell Lung Cancer	KRAS	p.G12C	248 (48.4%)
Non-Small Cell Lung Cancer	EGFR	p.L858R	74 (14.5%)
Non-Small Cell Lung Cancer	EGFR	p.E746_A750del	62 (12.1%)
Non-Small Cell Lung Cancer	BRAF	p.V600E	15 (2.9%)
Non-Small Cell Lung Cancer	MET	p.X1010_splice	13 (2.5%)
Non-Small Cell Lung Cancer	MET	p.X963_splice	11 (2.1%)
Non-Small Cell Lung Cancer	EGFR	p.E746_S752delinsV	8 (1.6%)
Non-Small Cell Lung Cancer	EGFR	p.L747_P753delinsS	7 (1.4%)
Non-Small Cell Lung Cancer	ERBB2	p.Y772_A775dup	7 (1.4%)
Non-Small Cell Lung Cancer	EGFR	p.G719A	5 (1%)
Oligodendroglioma	IDH1	p.R132H	406 (86.9%)
Oligodendroglioma	IDH2	p.R172K	41 (8.8%)
Oligodendroglioma	IDH2	p.R172M	7 (1.5%)
Oligodendroglioma	IDH1	p.R132S	4 (0.9%)
Oligodendroglioma	IDH1	p.R132G	3 (0.6%)
Oligodendroglioma	IDH2	p.R172W	3 (0.6%)
Oligodendroglioma	IDH1	p.R132C	2 (0.4%)
Oligodendroglioma	IDH2	p.R172S	1 (0.2%)
High-Grade Serous Ovarian Cancer	BRCA1	p.Q1756Pfs*74	19 (4.3%)
High-Grade Serous Ovarian Cancer	BRCA1	p.E23Vfs*17	15 (3.4%)
High-Grade Serous Ovarian Cancer	BRCA2	p.A938Pfs*21	5 (1.1%)
High-Grade Serous Ovarian Cancer	BRCA1	p.C61G	4 (0.9%)
High-Grade Serous Ovarian Cancer	BRCA1	p.R1699W	4 (0.9%)
High-Grade Serous Ovarian Cancer	BRCA1	p.X1559_splice	4 (0.9%)
High-Grade Serous Ovarian Cancer	BRCA2	p.L2357Vfs*2	4 (0.9%)
High-Grade Serous Ovarian Cancer	BRCA1	p.E1161Ffs*3	3 (0.7%)
High-Grade Serous Ovarian Cancer	BRCA1	p.E1250*	3 (0.7%)
High-Grade Serous Ovarian Cancer	BRCA1	p.K679*	3 (0.7%)
Astrocytoma	IDH1	p.R132H	259 (70.4%)
Astrocytoma	IDH1	p.R132C	33 (9%)
Astrocytoma	BRAF	p.V600E	30 (8.2%)
Astrocytoma	IDH1	p.R132G	19 (5.2%)
Astrocytoma	IDH1	p.R132S	14 (3.8%)
Astrocytoma	IDH2	p.R172G	5 (1.4%)
Astrocytoma	IDH2	p.R172M	4 (1.1%)
Astrocytoma	IDH1	p.R132L	2 (0.5%)
Astrocytoma	IDH2	p.R172K	1 (0.3%)
Astrocytoma	IDH2	p.R172S	1 (0.3%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.H1047R	120 (39.2%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.E545K	53 (17.3%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.E542K	28 (9.2%)
Breast Mixed Ductal and Lobular Carcinoma	AKT1	p.E17K	23 (7.5%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.N345K	14 (4.6%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.H1047L	12 (3.9%)
Breast Mixed Ductal and Lobular Carcinoma	ESR1	p.D538G	7 (2.3%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.C420R	7 (2.3%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.M1043I	5 (1.6%)
Breast Mixed Ductal and Lobular Carcinoma	PIK3CA	p.Q546K	5 (1.6%)